Managing Hyponatremia:

Fast or Slow? Why, When, How, and Controversies

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Abstract

Hyponatremia is the most common electrolyte disorder encountered in clinical practice. Its management is challenging because treatment must balance the risk of cerebral edema in acute hyponatremia against the risk of osmotic demyelination syndrome (ODS) from overly rapid correction of chronic hyponatremia. Recent observational studies have questioned current correction limits, prompting renewed debate regarding optimal management. This narrative review presents a physiology-based approach to the evaluation and treatment of hyponatremia. Acute hyponatremia may cause life-threatening cerebral edema because cerebral adaptation is incomplete, whereas chronic hyponatremia is characterized by the loss of intracellular electrolytes and organic osmolytes that limit brain swelling but increase vulnerability to osmotic injury during correction. Because the duration of hyponatremia is often uncertain, management should be guided primarily by neurological manifestations rather than duration alone. We propose a systematic treatment approach based on neurological severity, the duration of hyponatremia when definitively known, and the patient's risk of neurological injury, balancing the immediate danger of cerebral edema against the risk of ODS. We critically review the evidence underlying current correction limits and recent observational studies reporting that ODS is uncommon, inconsistently associated with overly rapid correction, and that slower correction has been associated with higher mortality. We conclude that, until higher-quality evidence becomes available, conservative correction limits for management of chronic hyponatremia remain appropriate, with correction not exceeding 10 mmol/L in 24 hours (18 mmol/L in 48 hours) in average-risk patients, targeting a rise of 4–8 mmol/L/day, and 8 mmol/L in 24 hours in patients at high risk for ODS, targeting a rise of 4–6 mmol/L/day.

Keywords: aldosterone, guidelines, hyponatremia, systematic review, vasopressin

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Hyponatremia, defined as a plasma sodium concentration (P_{Na}) < 135 mmol/L, is the most common electrolyte abnormality encountered in clinical practice [1], and its management can be challenging[2,3]. Acute hyponatremia can cause life-threatening cerebral edema and therefore requires prompt correction, whereas overly rapid correction of chronic hyponatremia may result in osmotic demyelination syndrome (ODS). Appropriate management requires an understanding of the physiology of water homeostasis, the pathophysiology of hyponatremia, the mechanisms of cerebral adaptation to hypotonic stress, and the pathophysiology of ODS[4]. Because the duration of hyponatremia is often uncertain, classification as acute or chronic may be unreliable. Initial management should therefore be guided by the presence and severity of neurologic symptoms suggestive of increased intracranial pressure, while subsequent correction should be based on the patient's risk for ODS. This review outlines an approach to the management of hyponatremia, examines the rationale underlying current recommendations for limits of correction of chronic hyponatremia, and discusses ongoing controversies regarding these limits.

Determinants of Water Movement across Cell Membranes

Water moves freely across cell membranes through aquaporin (AQP) channels, resulting in equal concentrations of effective osmoles in the extracellular fluid (ECF) and the intracellular fluid (ICF) compartments[5]. Effective osmoles are solutes that are largely confined to a single compartment and therefore generate osmotic forces that drive water movement across cell membranes. In the ECF, the principal effective osmoles are sodium (Na^+) ions and their accompanying anions, primarily chloride and bicarbonate. In the ICF, the principal effective osmoles are potassium (K^+) ions, organic phosphate anions, and organic osmolytes, including

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sugar derivatives (myoinositol and sorbitol), amino acids (taurine, glutamate, glutamine, glycine), and methylamines (betaine , glycerophosphorylcholine) [6].

In contrast, urea and ethanol readily cross cell membranes and rapidly equilibrate between the ECF and ICF. Although they measured plasma osmolality, they do not increase effective osmolality (tonicity) and therefore do not drive water movement across cell membranes[7]. Under normal conditions, insulin promotes glucose uptake into skeletal muscle by facilitating the insertion of GLUT4 transporters into the cell membrane; consequently, glucose does not function as an effective osmole in muscle. However, in states of absolute or relative insulin deficiency, glucose remains largely confined to the ECF, increasing its effective osmolality and drawing water out of muscle cells. This resulting shift of water from the ICF to the ECF produces dilutional hyponatremia despite an elevated plasma effective osmolality[8].

The intracellular content of effective osmoles is maintained relatively constant. Because Na^+ ions and their accompanying anions are the principal effective osmoles in the ECF, the plasma Na^+ concentration is the major determinant of ICF volume. Hyponatremia, whether due to Na^+ loss or water gain, lowers extracellular tonicity and promotes water movement into cells, resulting in cellular swelling. In response to extracellular hypotonicity, brain cells adapt by extruding electrolytes and organic osmolytes, thereby reducing intracellular osmolality and promoting water efflux to restore cell volume[9].

Physiology of Water Homeostasis

Water homeostasis maintains plasma osmolality (P_{Osm}) near 290 mOsm/kg H_2O, although normal values vary among individuals and generally range from 280–295 mOsm/kg H_2O

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[10].Primary osmosensory neurons are located in the organum vasculosum of the lamina terminalis and the subfornical organ, hypothalamic structures adjacent to the third ventricle. Because these regions lack a blood–brain barrier, they directly sense changes in plasma Na^+ concentration (P_{Na}) [11,12]. These neurons project to the median preoptic nucleus, where osmotic and hormonal signals are integrated [13] . Signals are then transmitted to the supraoptic and paraventricular nuclei of the hypothalamus, where arginine vasopressin (AVP) is synthesized [14], and to cortical regions, particularly the anterior cingulate cortex and insula, where thirst perception arises [15].

The AVP gene encodes a precursor prohormone that undergoes post-translational processing into three products: the biologically active nine–amino acid AVP peptide, neurophysin II (a carrier protein), and copeptin (a glycopeptide). These products are co-packaged in neurosecretory vesicles and transported along axons in the hypothalamic–hypophyseal tract to the posterior pituitary, where they are stored until release is stimulated [16].

AVP secretion is regulated primarily by P_{Osm}. An increase of approximately 1% above the osmotic threshold triggers a steep rise in circulating AVP levels[16,17]. Reductions in effective arterial blood volume (EABV) or arterial blood pressure also stimulate AVP release via baroreceptors located in the carotid sinus and aortic arch [18]. Afferent signals from these receptors travel through the glossopharyngeal and vagus nerves to the nucleus tractus solitarius in the brainstem and are subsequently relayed to the hypothalamus, where they modulate AVP release from the posterior pituitary. Under euvolemic conditions, tonic baroreceptor input suppresses AVP secretion. In humans, plasma volume reductions have little effect on AVP levels until volume depletion reaches approximately 10–15%. Larger reductions (20–30%) produce marked increases in AVP concentrations, far exceeding those required for maximal antidiuresis[18,19] .

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Non-osmotic stimuli for AVP secretion include nausea, pain, physical and psychological stress, inflammatory cytokines, medications including opiates, ifosfamide, and vincristine, and recreational drugs such as 3,4-methylenedioxymethamphetamine (MDMA) [20,21] .

AVP acts on vasopressin V2 receptors (AVPR2), which are G protein–coupled receptors located on the basolateral membrane of collecting duct principal cells. Activation of these receptors stimulates adenylyl cyclase, increasing intracellular cyclic adenosine monophosphate and activating protein kinase A (PKA). PKA phosphorylates aquaporin-2 (AQP2) water channels stored within intracellular vesicles, promoting their trafficking and insertion into the luminal membrane. Insertion of AQP2 into the luminal membrane renders principal cells highlyermeable to water, permitting water reabsorption until the effective osmolality of the tubular fluid equilibrates with that of the surrounding interstitium [22,23].

Pathophysiology of Hyponatremia

Because water moves freely across cell membranes, osmolality remains equal in the ECF and ICF compartments. Accordingly, the concentration of effective osmoles must be equal in the ECF and ICF. This establishes a relationship between P_{Na}, total body Na^+ and K^+ content, and total body water (TBW). Using isotope dilution techniques, Edelman and colleagues described this relationship as [24] :

$$[Na^+]_{\text{plasma water}} = 1.11 \times (Na_e^+ + K_e^+)/TBW - 25.6 \quad (1)$$

where Na_e^+ and K_e^+ denote exchangeable sodium and potassium. The slope of the regression (1.11) reflects the osmotic coefficient of sodium salts and the Gibbs–Donnan equilibrium, whereas the negative intercept reflects the presence of osmotically inactive stores of

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exchangeable Na^+ and K^+. Na^+ may be stored in an osmotically inactive form bound to negatively charged glycosaminoglycans within the skin and muscle interstitium [25]. Whether dynamic activation or inactivation of these Na^+ stores significantly influences P_{Na} levels remains uncertain[26].

For clinical application, the equation is simplified by omitting the intercept, substituting total body sodium and potassium for their exchangeable fractions, and using the measured plasma Na^+ concentration:

P_{Na} = total body $(Na^+ + K^+)$/TBW (2)

Hyponatremia therefore occurs when TBW is increased relative to total body Na^+ and K^+, assuming no other effective osmoles are present in the ECF. This imbalance may result from isolated water gain, loss of Na^+ and K^+, combined solute and water losses with proportionally greater solute loss, or concurrent gains of both solute and water when water gain exceeds solute gain.

Acute hyponatremia most commonly results from ingestion of large amounts of electrolyte-free water in the setting of impaired free water excretion due to non-osmotic AVP release. Common clinical settings include the perioperative state [27]; absorption of hypotonic irrigation fluid during procedures such as transurethral resection of the prostate or hysteroscopic transcervical endometrial resection, [28] psychogenic polydipsia, [29] exercise-associated hyponatremia, [30] and colonoscopy preparation [31].

Chronic hyponatremia most often results from impaired electrolyte-free water excretion due to persistent AVP activity. In such cases, AVP secretion may be physiologically appropriate, as occurs with reduced EABV in true hypovolemia or hypervolemic states such as advanced heart

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failure or cirrhosis, or inappropriate despite normal EABV, a condition traditionally known as the syndrome of inappropriate antidiuretic hormone secretion (SIADH)[20,32,33].

SIADH may occur in association with malignancy, particularly small-cell lung carcinoma; medications, including SSRIs; pulmonary diseases such as bacterial, viral, or tuberculous pneumonia; and central nervous system disorders including tumors, infections, cerebrovascular events, trauma, and pituitary surgery. Rarely, constitutive activation of the V2 receptor produces nephrogenic syndrome of inappropriate antidiuresis independent of AVP. The broader term "syndrome of inappropriate antidiuresis" (SIAD) encompasses both central (AVP-dependent) and nephrogenic (AVP-independent) forms[34].

SIAD is a diagnosis of exclusion and requires ruling out cortisol deficiency and severe hypothyroidism. Importantly, some causes of hyponatremia including thiazide-induced hyponatremia[35] , low-solute intake ("tea and toast" hyponatremia) [36] and beer potomania [37],are not typically associated with sufficiently reduced EABV to trigger baroreceptor-mediated AVP release. In these cases, impaired electrolyte-free water excretion may result from markedly reduced distal delivery of filtrate together with increased water reabsorption in the inner medullary collecting duct[38]. Reduced distal delivery may result from a decrease in glomerular filtration rate (GFR) and enhanced proximal tubular reabsorption of Na^+ mediated by sympathetic nervous system activation and angiotensin II release in response to relatively modest reductions in EABV. This mechanism may contribute to hyponatremia in patients with ischemic kidney disease and reduced GFR who are receiving thiazide diuretics[35]. The inner medullary collecting duct retains low but measurable water permeability even in the absence of AVP, probably reflecting a small basal expression of AQP2 in the luminal membrane [39,40].When solute

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excretion is markedly reduced, as occurs with low salt and protein-intake, a steep osmotic gradient develops between the tubular lumen and the medullary interstitium, promoting water reabsorption and contributing to hyponatremia. Accordingly, conditions associated with impaired electrolyte-free water excretion due to reduced distal delivery of filtrate and increased water reabsorption through residual water permeability in the inner medullary collecting duct should be excluded before a diagnosis of SIAD is established.

Brain Adaptation to Hyponatremia

Regulation of brain cell volume is critical because the brain is enclosed within a rigid skull; consequently, even modest increases in brain volume can raise intracranial pressure (ICP). When extracellular hypotonicity develops, the brain initiates regulatory volume-decrease mechanisms to restore cell volume toward normal [41,42].

The initial response involves displacement of interstitial and cerebrospinal fluid from the cranial vault to the spinal subarachnoid space and systemic circulation. This is followed rapidly by loss of intracellular osmoles, resulting in water efflux and reduction of cellular swelling. Experimental studies in rats with acute dilutional hyponatremia demonstrate that Na^+ and Cl^- losses begin within 30 minutes of the onset of hyponatremia with K^+ losses occurring slightly later. Electrolyte losses peak after approximately three hours but are limited to roughly 18% of the total brain electrolyte content[43,44] . In prolonged hyponatremia (>48 hours), slower adaptive mechanisms develop, characterized by loss of organic osmolytes, including glutamate, glutamine, taurine, myo-inositol, creatine, and glycerophosphorylcholine. These processes restore brain water to near normal levels. [45,46] Approximately two-thirds of this volume regulation results from electrolyte loss and one-third from depletion of organic osmolytes [45].

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Because many of these osmolytes play important roles in neurotransmission and protein folding [47,48], their depletion may contribute to the cognitive impairment, gait disturbances, and increased risk of falls observed in patients with chronic hyponatremia.

When hyponatremia develops rapidly, these adaptive mechanisms do not have sufficient time to occur, and even moderate P_{Na} reductions can result in significant cerebral edema. Rising ICP reduces cerebral perfusion pressure and may lead to brain hypoxia. Downward displacement of the brain may compress cerebral veins at the foramen magnum, impairing venous drainage and exacerbating edema. Non-cardiogenic pulmonary edema may develop and further exacerbate cerebral hypoxia and brain swelling. Severe neurological complications including seizures, coma, brain herniation, and death may ensue [8].

During correction of chronic hyponatremia (>48 hours), brain electrolyte content is typically restored within 24 hours and may transiently overshoot baseline levels. In contrast, reaccumulation of organic osmolytes is considerably slower, often requiring several days, owing to downregulation of transporters and synthetic pathways during the adaptation to hyponatremia [46,49,50]. This lag creates osmotic stress when P_{Na} rises rapidly and is believed to be a central mechanism predisposing to ODS.

Osmotic Demyelination Syndrome (ODS)

Central pontine myelinolysis (CPM) was originally described in patients with chronic alcoholism but is now recognized to occur in association with overly rapid correction of chronic hyponatremia. CPM is characterized by symmetrical, noninflammatory demyelination of the central pons with relative preservation of neurons. Experimental studies demonstrate that demyelination occurs primarily when chronic hyponatremia of more than 48–72 hours' duration

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is corrected too rapidly, whereas both acute and untreated chronic hyponatremia rarely produce such lesions[51-53]. Animal studies further suggest that re-lowering P_{Na} within 12 hours after overly rapid correction can reduce demyelination and mortality[54]. The term osmotic demyelination syndrome (ODS) was introduced to emphasize the osmotic link and to reflect the involvement of regions beyond the pons, including the basal ganglia, thalamus, cerebellum, and subcortical white matter, a condition termed extrapontine myelinolysis (EPM)) [55].

The pathophysiology of ODS remains incompletely understood but astrocyte injury appears to be an early and central event [56]. Astrocytes, whose AQP4–rich endfeet envelop cerebral capillaries, play a critical role in protecting neurons from osmotic stress and maintaining brain volume homeostasis [42]. During hypotonic conditions, neurons release osmolytes such as taurine into the extracellular space, which are subsequently taken up by astrocytes, leading them to swell. Within 24–48 hours, astrocytes undergo regulatory volume decrease, releasing osmolytes and ions including K^+ and Cl^- into the extracellular space [57,58]. When hyponatremia is corrected rapidly, these osmolyte-depleted astrocytes are exposed to abrupt osmotic stress, causing cell shrinkage, and resulting in disruption of protein folding within the endoplasmic reticulum and activation of the unfolded protein response. Severe or sustained stress may lead to astrocyte death through apoptotic and necroptotic pathways [59]. Because oligodendrocytes depend on astrocytes for trophic support, secondary oligodendrocyte injury may ensue, culminating in demyelination.

Another proposed mechanism for ODS involves a rapid increase in P_{Na} leading to shrinkage of cerebral endothelial cells, disruption of the blood–brain barrier, and entry of inflammatory mediators into the brain parenchyma, resulting in oligodendrocyte injury [60]. However,

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experimental data suggest that astrocyte injury precedes both demyelination and blood–brain barrier disruption [61]. Microglial activation and inflammatory cytokine release may further exacerbate oligodendrocyte damage [62].

Clinically, CPM may present with dysarthria, dysphagia, pseudobulbar palsy, pupillary and oculomotor abnormalities, and a flaccid quadriparesis that evolves into spasticity. Extensive pontine demyelination can result in the classic "locked-in syndrome," characterized by quadriplegia and anarthria with preserved consciousness and vertical eye movements. Extrapontine lesions may manifest as altered mental status, behavioral disturbances, seizures, or movement disorders.[63]

The diagnosis of ODS can be challenging. Its clinical course is often biphasic, with initial neurological improvement after partial correction of $P_{Na,}$ followed days later by the development of new neurological deficits. Manifestations vary according to the location and extent of demyelination[42,64]. Magnetic resonance imaging is more sensitive than computed tomography in detecting lesions consistent with ODS; however early imaging may be normal, and the characteristic T2-weighted hyperintense lesions may not become apparent until 2- 4 weeks after symptom onset [65].

The risk of ODS is not determined solely by the rate of correction. Susceptibility varies substantially among patients and is greatest in those with very low baseline PNa (particularly ≤105–110 mmol/L), alcohol use disorder, advanced liver disease, malnutrition, and hypokalemia . In contrast, ODS is uncommon in patients without these risk factors, particularly when the initial P_{Na} is above 120 mmol/L. Accordingly, recommended correction limits should be individualized according to the patient's baseline risk of ODS.

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Management of Hyponatremia

The principal challenge in the management of hyponatremia is balancing two competing risks: preventing cerebral edema and brain herniation in acute hyponatremia while avoiding overly rapid correction which may precipitate ODS in chronic hyponatremia.

Even moderate declines in P_{Na} can be life-threatening when they occur rapidly. Patients may initially present with mild or nonspecific symptoms such as mild headache, mild nausea, fatigue, lethargy, or mild cognitive impairment, but can quickly deteriorate to obtundation, seizures, respiratory distress, coma, or death. This rapid deterioration has been particularly observed in settings such as postoperative states, exercise-associated hyponatremia, psychogenic polydipsia, or MDMA use [42,64].

Patients with chronic hyponatremia face a dual risk: further acute decreases in P_{Na} can increase ICP, whereas overly rapid correction may lead to ODS.

Because the duration of hyponatremia is often uncertain, classification as acute or chronic may be unreliable. Accordingly, management should be guided primarily by the presence and severity of neurologic symptoms, which reflect increased intracranial pressure. The therapeutic goal is to raise P_{Na} enough to reduce cerebral edema and prevent brain herniation while remaining within safe correction limits to minimize the risk of ODS, once any immediate neurologic risk has been addressed.

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i) Patients with Severe Symptoms:

Symptoms such as seizures, obtundation or coma, vomiting, and respiratory distress are alarming for cerebral edema and increased ICP. In patients with such symptoms, the immediate therapeutic goal is prompt reduction of ICP. The treatment of choice is administration of hypertonic saline (3% NaCl) to rapidly increase the P_{Na} (Flow Chart 1). Although the blood–brain barrier restricts passive Na^+ movement, hypertonic saline increases P_{Osm}, creating an osmotic gradient that shifts water from brain tissue into the intravascular compartment. Water flux is mediated primarily through AQP4 channels located in astrocytic endfeet [66].

Evidence from neurosurgical patients with elevated ICP indicates that increasing P_{Na} by approximately 5 mmol/L can substantially reduce intracranial pressure and reverse impending brain herniation, even when baseline P_{Na} levels are normal [67-69]. Based on these observations and clinical experience in endurance athletes with acute hyponatremia, both United States and European guidelines recommend raising the P_{Na} by approximately 4–6 mmol/L within the first 1–2 hours [42,64]. This degree of correction is generally sufficient to reverse life-threatening cerebral edema and neurological symptoms. The target is typically achieved by administering 100–150 mL of 3% hypertonic saline intravenously over 10–20 minutes, repeated up to two or three times as necessary to achieve a 4-6 mmol/L increase in P_{Na} [70,71]. Hypertonic saline can be administered safely through a peripheral vein, allowing treatment to be initiated without delay [72].

An individualized dosing strategy for hypertonic saline may be used based on estimated TBW. Although the infused Na^+ remains within the ECF, its osmotic effect distributes across TBW. Because 1 liter of 3% saline contains 513 mmol of Na^+ (approximately 1 mmol per 2 mL), the required volume can be estimated as follows:

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Volume of 3% saline ≈ desired rise in P_{Na} × TBW × 2.

Assuming TBW is about 50% of body weight, this simplifies to:

Volume of 3% saline ≈ desired rise in P_{Na} × body weight (kg).

The estimated volume may be increased by about 10% in lean individuals and decreased by about 10% in obese individuals. However, administration of 3% NaCl provides both NaCl and water. The administered water slightly blunts the increase in P_{Na} that would be achieved by the added Na^+ alone. To account for this effect, an additional volume of 3% NaCl may be required, generally about 25% above the initially calculated volume. This formula, however, has not been clinically validated, regardless of the initial calculated dose, therapy must be guided by frequent reassessment of P_{Na}, with subsequent adjustments in hypertonic saline administration based on the measured change in P_{Na}.

If severe symptoms persist after the initial P_{Na} increase of 5 mmol/L and the duration of hyponatremia is clearly < 48 hours, European guidelines recommend additional hypertonic saline to achieve a further increase of 5 mmol/L[71,73].

If symptoms resolve after the initial 4-6 mmol/L rise and hyponatremia duration is clearly < 48 hours, hypertonic saline may be continued to restore the P_{Na} toward the normal range over several hours. However, if hyponatremia has been present for more than 24 hours, correction should generally not exceed 10 mmol/L within 24 hours. Exceeding this limit is unlikely to provide additional clinical benefit, and although the risk of ODS is very low in such patients, it may not be entirely absent, depending on the degree of cerebral adaptation.

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When the duration is uncertain, correction in the first 24 hours and each subsequent 24-hour period should remain within the recommended limits based on patient's ODS risk (Flow Chart 1). correction not exceeding 10 mmol/L in 24 hours (18 mmol/L in 48 hours) in average-risk patients, targeting a rise of 4–8 mmol/L/day, and 8 mmol/L in 24 hours in patients at high risk for ODS, targeting a rise of 4–6 mmol/L/day

ii) Patients with Moderately Severe Symptoms:

Moderately severe symptoms of hyponatremia include nausea without vomiting, headache, and confusion, or disorientation.

When the duration of hyponatremia is known to be less than 48 hours, treatment is similar to that recommended for patients with severe symptoms (Flow Chart 2).

When the duration of hyponatremia is uncertain, management requires careful consideration. Symptoms may result from an acute reduction in P_{Na} causing increased ICP in a patient with chronic hyponatremia, from severe chronic hyponatremia itself (typically P_{Na} <120 mmol/L), due to changes in brain cell composition associated with cerebral adaptation to chronic hypotonicity[74,75] , or from conditions unrelated to hyponatremia. Management should therefore be guided by the likelihood that symptoms reflect increased ICP and the patient's risk of neurologic injury.

Patients with an extremely low P_{Na} (e.g., <110 mmol/L); a history suggestive of a recent acute decline in P_{Na} (e.g., excessive water intake), or conditions that increase the risk for neurologic complications, including intracranial disease or seizure disorders, should be managed as though increased ICP is present. In such patients, hypertonic saline should be administered with the goal

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of increasing P_{Na} by approximately4-6 mmol /L over 1 to 2 hours. Close monitoring is required to avoid overcorrection, particularly if a spontaneous water diuresis develops. Thereafter, the overall increase in P_{Na} during the first 24 hours and each subsequent 24-hour period should not exceed the recommended correction limits based on the patient's risk for ODS.

In patients without these high-risk features, for whom the presence of increased ICP remains uncertain, management depends on whether water diuresis is present. If the patient is undergoing water diuresis, spontaneous correction is allowed to continue, until P_{Na} rises by approximately 5 mmol/L, after which DDAVP should be administered to prevent overcorrection. In the absence of water diuresis, 3% hypertonic saline may be infused at a controlled rate to increase P_{Na} by 0.5 to 1 mmol/L per hour, while limiting the total increase to no more than 5 mmol/L. Thereafter, the overall increase in P_{Na} during the first 24 hours and each subsequent 24-hour period should not exceed the recommended correction limits based on the patient's risk for ODS.

For patients whose symptoms are not believed to reflect increased ICP, a more conservative strategy is appropriate. In such cases, the goal is gradual correction without exceeding the the recommended correction limits based on the patient's risk for ODS .

iii) Patients with No Severe and no Moderately Severe Symptoms:

Management in these patients depends on whether the duration of hyponatremia is known with certainty to be less than 48 hours (e.g., postoperative hyponatremia) or whether the duration is uncertain.

a) Patients with a known duration of hyponatremia of less than 48 hours

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It may be argued that all patients with acute hyponatremia should receive intravenous hypertonic saline because even seemingly nonspecific symptoms, such as mild headache, nausea, fatigue, lethargy, or subtle cognitive impairment, may precede rapid neurologic deterioration. However, such an approach may not be universally practical, particularly if it requires admission to a monitored or critical care setting. A more practical strategy is to identify patients at risk for a further rapid decline in P_{Na} and those who may be especially vulnerable to neurologic injury. In these higher-risk patients, administration of intravenous hypertonic saline is warranted.

A sudden decline in P_{Na} may occur when large volumes of ingested water are transiently retained in the stomach and then rapidly delivered to the small intestine and absorbed. In such cases, the brain may be exposed to an abrupt fall in arterial P_{Na}, before changes are detected in venous blood samples [76]. The P_{Na} may also decline when isotonic saline is administered and the salt load is excreted in hypertonic urine due persistent non-osmotic AVP release, leaving behind electrolyte-free water in the body , a phenomenon termed "desalination" [77].Patients with low muscle mass are particularly susceptible to large decreases in P_{Na} because relatively small volumes of electrolyte-free water produce proportionally greater increases in TBW.

Certain populations are especially vulnerable to neurologic injury. Menstruating women with acute hyponatremia may be at increased risk [78], owing to the combined effect of estrogen and AVP inhibiting the Na^+–K^+-ATPase activity and promoting cerebral vasoconstriction, thereby impairing brain adaptation to acute hypotonicity and predisposing to cerebral hypoxia [79]. Patients with underlying intracranial pathology, including recent traumatic brain injury, neurosurgery, intracranial hemorrhage, or tumors, as well as those with seizure disorders are at increased risk of neurological complications with relatively modest increases in ICP

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In patients considered to be at high-risk for either rapid decline in P_{Na} or neurological injury, hypertonic saline should be administered to increase P_{Na} by 4-6 mmol /L at a rate of about 1 mmol /L/hour. Correction may then proceed over several hours to raise P_{Na} toward the normal range. In patients whose hyponatremia has been present for more than 24 hours, the increase in P_{Na} should not exceed 10 mmol/L within 24 hours. Close monitoring is required, with adjustment of therapy if spontaneous water diuresis develops.

Patients with known duration of hyponatremia of less than 48 hours who do not exhibit severe or moderately severe symptoms and who are not considered at risk for rapid declines in P_{Na} or neurologic injury, may be managed more conservatively with fluid restriction and adjunctive therapies, including oral sodium chloride, loop diuretics, urea, or vasopressin receptor antagonists, under close monitoring. The goal is gradual correction to raise the P_{Na} toward the normal range while limiting the increase to no more than 10 mmol/L within 24 hours when hyponatremia has been present for more than 24 hours.

b) Patients with known duration of hyponatremia greater than 48 hours or hyponatremia of uncertain duration

The principal therapeutic concern is prevention of ODS resulting from overly rapid correction of hyponatremia. Although chronic hyponatremia is often described as asymptomatic, patients may exhibit mild cognitive impairment, gait disturbances, and an increased risk of falls.

Rapid correction most commonly occurs when the underlying cause of impaired water excretion is reversed. Examples include correction of hypovolemia, glucocorticoid replacement in patients with cortisol deficiency, resolution of transient stimuli for AVP release (e.g., pain, nausea,

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infections, medications such as SSRIs), or discontinuation of DDAVP. Because these events may lead to abrupt increases in P_{Na}, DDAVP may be used prophylactically or reactively to prevent excessive water diuresis and overcorrection [80,81].

Retrospective studies have demonstrated an association between rapid correction of severe chronic hyponatremia (P_{Na} <120 mmol/L) and ODS [74,82-87]. However, these studies have important limitations, including small sample sizes, inconsistent timing of P_{Na} measurements used to estimate correction rates, variable definitions of rapid correction, emphasis on changes during the first 24 hours, and heterogeneous criteria for case ascertainment. Although early case series suggested a high incidence of ODS after rapid correction, larger contemporary studies have reported substantially lower rates [86,87].Importantly, most patients who undergo rapid correction do not develop ODS, indicating that additional factors beyond the rate of correction influence susceptibility to ODS. The risk appears greatest in patients with very low initial P_{Na} (≤ 105–110 mmol /L). The risk is also increased in patients with alcohol use disorder, advanced liver disease, malnutrition, hypokalemia, or hypophosphatemia, in whom ODS has been reported even when the initial P_{Na} exceeded 120 mmol/L.

Although there is broad consensus that overly rapid correction of chronic hyponatremia can precipitate ODS, recommended correction limits differ among guidelines.

The United States/Irish expert panel recommends that in patients at average risk of ODS, P_{Na} correction should not exceed 10–12 mmol/L in any 24-hour period or 18 mmol/L in 48 hours, with a minimum correction goal of 4–8 mmol/L per 24 hours. In high-risk patients, correction should not exceed 8 mmol/L in 24 hours, with a recommended minimum of 4–6 mmol/L during that period [70]. The European Clinical Practice Guidelines recommend a maximum increase in

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PNa of 10 mmol/L during the first 24 hours and 8 mmol/L during each subsequent 24-hour period, with a stricter limit of 8 mmol/L per 24 hours from the outset in patients at high risk for ODS. However, unlike the US/Irish Expert Panel, they do not specify a minimum target for the daily increase in P_{Na} [71]. Some experts advocate more conservative limits than those recommended by the United States/Irish Expert Panel and European Clinical Practice Guidelines. They recommend limiting correction to no more than 8 mmol /L within 24 hours for average-risk patients and no more than 6 mmol /L within 24 hours in high-risk patients, with an increase of 4–6 mmol /L in 24 hours considered an adequate therapeutic target [64,88,89]. The rationale for these stricter limits is that an increase of 4–6 mmol /L is sufficient to reduce ICP, whereas more rapid correction has not been shown to confer additional clinical benefit. Furthermore, although ODS is very uncommon, cases have been reported despite correction of P_{Na} by only 8–10 mmol/L within 24 hours, particularly in patients with an initial P_{Na} below 115 mmol /L [86,87,90].

Design of therapy for patients with chronic or uncertain duration of hyponatremia

Asymptomatic patients with P_{Na} >120 mmol/L who lack high-risk features for ODS can often be managed as outpatients. Management includes discontinuation of medications associated with hyponatremia, most notably thiazide diuretics and SSRIs, restriction of water intake, and serial monitoring of P_{Na} [91].

Patients with a P_{Na} below 120 mmol /L or those at high risk for ODS, require cautious correction, targeting an increase in P_{Na} of approximately 4–6 mmol /L per day while remaining within the limits described above.

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In absence of clinical hypervolemia, hyponatremia generally reflects a deficit of Na^+ in the ECF compartment and an excess water within the intracellular compartment, even when of overt ECF volume depletion is not clinically apparent. Initial management includes discontinuation of offending medications, restriction of fluid intake to approximately 800 ml per day and gradually replacing the sodium deficit through dietary sodium or divided doses of oral sodium chloride. The development of a water diuresis indicates restoration of the EABV and increased glomerular filtrate delivery to the distal nephron. At this stage, Na^+ supplementation is discontinued, and further correction is achieved through controlled negative water balance. DDAVP may be used to prevent excessive water diuresis and overly rapid correction. Correction of concomitant hypokalemia with potassium chloride may also increase P_{Na}, necessitating careful monitoring because patients with chronic hyponatremia and hypokalemia are at increased risk for ODS[92,93]..

An alternative strategy for managing chronic euvolemic hyponatremia involves scheduled administration of DDAVP to prevent spontaneous water diuresis while simultaneously administering 3% hypertonic saline at a calculated rate to achieve a controlled daily increase in P_{Na} [94]. This strategy allows tighter control of the correction rate but carries potential risks, including Na^+ overload, heart failure in susceptible patients, and recurrence of hyponatremia if the administered Na^+ load is later excreted in hypertonic urine while the effect of DDAVP persists. Despite these concerns, this strategy is particularly useful in patients at high risk for ODS when precise early control is required, at least until the P_{Na} reaches approximately 125 mmol/L.

In SIAD, persistent AVP activity results in retention of electrolyte-free water. Urine volume in this setting is determined by the rate of excretion of effective osmoles and medullary interstitial

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tonicity. Because AVP promotes insertion of urea transporters in the inner medullary collecting duct, urea equilibrates between the tubular fluid and interstitium and does not function as an effective osmole, Consequently, Na^+ and K^+ salts are the principal effective osmoles promoting water excretion.

Correcting hyponatremia in patients with SIAD requires generation of a negative electrolyte-free water balance. This may be achieved through fluid restriction, increasing effective osmole excretion, reducing medullary interstitial tonicity with a loop diuretic, or inducing water diuresis with V2 receptor antagonists (vaptans). Effective osmole excretion can be increased by increasing salt intake or the administration of urea at a rate that exceeds the capacity of its transport in the inner medullary collecting duct. Urea is typically administered at a dose of 30 g daily, providing approximately 500 mOsm of solute[95,96]. A low dose of loop diuretic (e.g., furosemide 20 mg) is usually sufficient to reduce medullary interstitial osmolality, twice-daily dosing is often required because of furosemide short duration of action [97]. Vaptans promote aquaresis through competitive antagonism of the V2 receptor and have been shown to increase P_{Na} in SIAD, heart failure, and cirrhosis[98,99]. Meta-analyses have reported rates of overcorrection ranging from 13% to 31%, depending on the threshold used to define overcorrection, even when low doses of tolvaptan are used for SIADH-related hyponatremia [100,101] . Nevertheless, no cases of ODS have been reported in clinical trials or meta-analyses. Because of concerns about hepatotoxicity, regulatory authorities have limited the use of tolvaptan to 30 days and have contraindicated its use in patients with underlying liver disease.

Emerging evidence suggests that SGLT2 inhibitors, particularly empagliflozin, may serve as adjuncts to fluid restriction in SIAD [102,103] .

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Controversy Regarding Hyponatremia Correction Rates

Recent observational studies have challenged the current recommendations for the rate of correction of chronic hyponatremia. These studies raise two important concerns. First, they suggest that ODS is very uncommon and may not be consistently associated with overly rapid correction of hyponatremia. Second, they propose that slower rates of correction of hyponatremia is associated with increased mortality.

In a large multicenter retrospective cohort study involving 22,858 hospitalized adults with P_{Na} <130 mmol/L, the mean initial P_{Na} was 125 mmol/L; only 11.9% had values between 110–119 mmol/L and 1.2% had values below 110 mmol/L[104] . Rapid correction, defined as increase in P_{Na} exceeding 8 mmol/L in 24 hours occurred in 17.7% of admissions. Despite this, confirmed or probable ODS, identified by neuroimaging or diagnostic coding, was rare, occurring in only 12 patients (0.05%). More than half of these cases occurred without documented rapid correction. Although, the incidence of ODS among patients with P_{Na} <110 mmol/L was higher (2.6%), the small number of events limited interpretation. An accompanying editorial argued that current guidelines may be overly restrictive, noting that they are based largely on low-quality evidence and may underestimate the potential harms of overly slow correction, including increased mortality and prolonged hospitalization [105].

In response, an international expert panel from 20 academic centers emphasized that current guideline limits for the rates of correction of hyponatremia were not intended to imply that exceeding them invariably results in ODS, but rather the risk increases in susceptible patients [106]. The panel highlighted limitations of the cohort study, including the predominance of patients with a P_{Na} above 120 mmol/L and the inclusion of disorders such as acute water intoxication and

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hyperglycemia-associated hyponatremia, both of which carry a low risk of ODS. The panel also noted that the reported 2.6% incidence of ODS among patients with a P_{Na} below 110 mmol/L may have underestimated its true frequency because ODS is primarily a clinical diagnosis, often presenting several days after correction and an MRI performed early in the course is often normal with radiological changes appearing later on or not at all. Although uncommon, ODS remains a devastating and potentially preventable complication of overly rapid correction of hyponatremia. Pending more definitive evidence, the panel urged clinicians to "stay the course" and maintain current conservative correction limits: no more than 10 mmol/L in 24 hours (or 18 mmol/L in 48 hours) for most patients, and no more than 8 mmol/L in 24 hours for patients at high risk, including those with initial P_{Na} ≤105 mmol/L, alcohol use disorder, malnutrition, hypokalemia, or advanced liver disease.

Additional evidence supports an association between CPM and rapid correction of hyponatremia in high-risk populations [107]. In a Swedish cohort of 83 patients with CPM, most had chronic hyponatremia (median P_{Na} ,104 mmol/L), and 70% had alcohol use disorder. The median correction rate was 17.3 mmol/L per 24 hours, substantially exceeding recommended limits; only six patients were corrected within guideline limits (≤ 8 mmol/L per 24 hours).

Further controversy arises from studies suggesting that slower correction may be associated with worse outcomes. In a multicenter cohort of 3,274 hospitalized patients with severe hyponatremia (P_{Na} ≤120 mmol/L), correction of less than 6 mmol/L in the first 24 hours was associated with higher in-hospital and 30-day mortality than correction of 6–10 mmol/L. Correction exceeding 10 mmol/L per 24 hours was associated with lower in-hospital mortality and shorter hospital stays. However, this apparent mortality benefit disappeared after propensity adjustment,

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suggesting residual confounding. ODS was rare, with only seven cases identified, most occurring in patients with recognized risk factors including alcohol use disorder, malnutrition, hypokalemia, or hypophosphatemia, and five of the seven cases occurred despite correction ≤8 mmol/L per 24 hours [108].

Similarly, a meta-analysis of 16 cohort studies involving 11,811 hospitalized adults with severe hyponatremia found that rapid correction (≥8–10 mEq/L per 24 hours) was associated with lower in-hospital mortality compared with slower correction, based on moderate-certainty evidence [109]. Rapid correction was also associated with shorter lengths of stay and lower 30-day mortality, although these findings were supported by low-certainty evidence. Rapid correction was not associated with a statistically significant increase in ODS, however, as acknowledged by the authors, the analysis may have been underpowered to detect such a rare complication.

These findings should be interpreted cautiously. None of these studies evaluated deaths attributable to cerebral edema, the principal mechanism by which insufficient correction might increase mortality[110]. Moreover, the association between faster correction and improved survival may reflect confounding by illness severity rather than a causal relationship [110,111] . Patients with advanced comorbid conditions, such as liver disease, heart failure, or cancer, often correct more slowly because of persistent AVP release and impaired renal free-water excretion. Consequently, improved outcomes observed with more rapid correction should not be interpreted as justification for exceeding current guideline limits.

Additional insights are provided from an emergency department cohort of 852 patients with severe hyponatremia (P_{Na} ≤125 mmol/L) [112]. Cerebral edema at presentation was documented in only 4 patients (0.5%), even though 37% of the cohort presented with severe neurologic

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symptoms. However, its incidence may have been underestimated because of the retrospective study design, the fact that fewer than half of symptomatic patients underwent neuroimaging, and the restrictive definition of cerebral edema, which was limited to diffuse global edema and excluded milder radiographic abnormalities. In contrast, ODS, defined as the development of new neurologic deficits after correction of P_{Na} together with radiologic evidence of demyelination, occurred in 1.3% of patients. Thus, the reported incidence of ODS exceeded that of cerebral edema, although the latter may have been underestimated. Nevertheless, the authors suggested that aggressive treatment may not be necessary in many patients with severe hyponatremia . The rate of P_{Na} correction was not associated with mortality or length of hospital stay. Although rapid correction (>8 mmol/L within 24 hours) was not associated with ODS in the primary analysis, a secondary analysis that used more accurate estimates of the 24-hour correction rate identified rapid correction as the principal risk factor for ODS, particularly in patients with sustained overcorrection and established risk factors. This discrepancy highlights an important limitation of retrospective studies: accurate determination of correction rates depends on the timing and frequency of $_{PNa}$ measurements and may therefore be subject to substantial measurement bias.

In summary, although ODS may be very uncommon, it remains a serious and potentially preventable complication of overly rapid correction of chronic hyponatremia. The apparent benefits of more rapid correction on mortality likely reflect confounding rather than causation. Until higher-quality evidence becomes available, current conservative correction limits for chronic hyponatremia remain appropriate, particularly in patients at increased risk of ODS. Accordingly, the increase in P_{Na} should not exceed 10 mmol/L in any 24-hour period (or 18 mmol/L in 48 hours) in most patients and 8 mmol/L in any 24-hour period in those at high risk

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for ODS, with a target increase of 4–8 mmol/L per day in average-risk patients and 4–6 mmol/L per day in high-risk patients.

Conflict of interest statement

Kamel S Kamel reports royalties from Elsevier for publication of the fifth edition of the textbook "Fluid and Electrolytes and Acid–Base Physiology: A Problem-Based Approach". Ziv Harel and Martin Schreiber report no conflict of interest.

Data availability statement

No new data were generated or analysed in support of this research.

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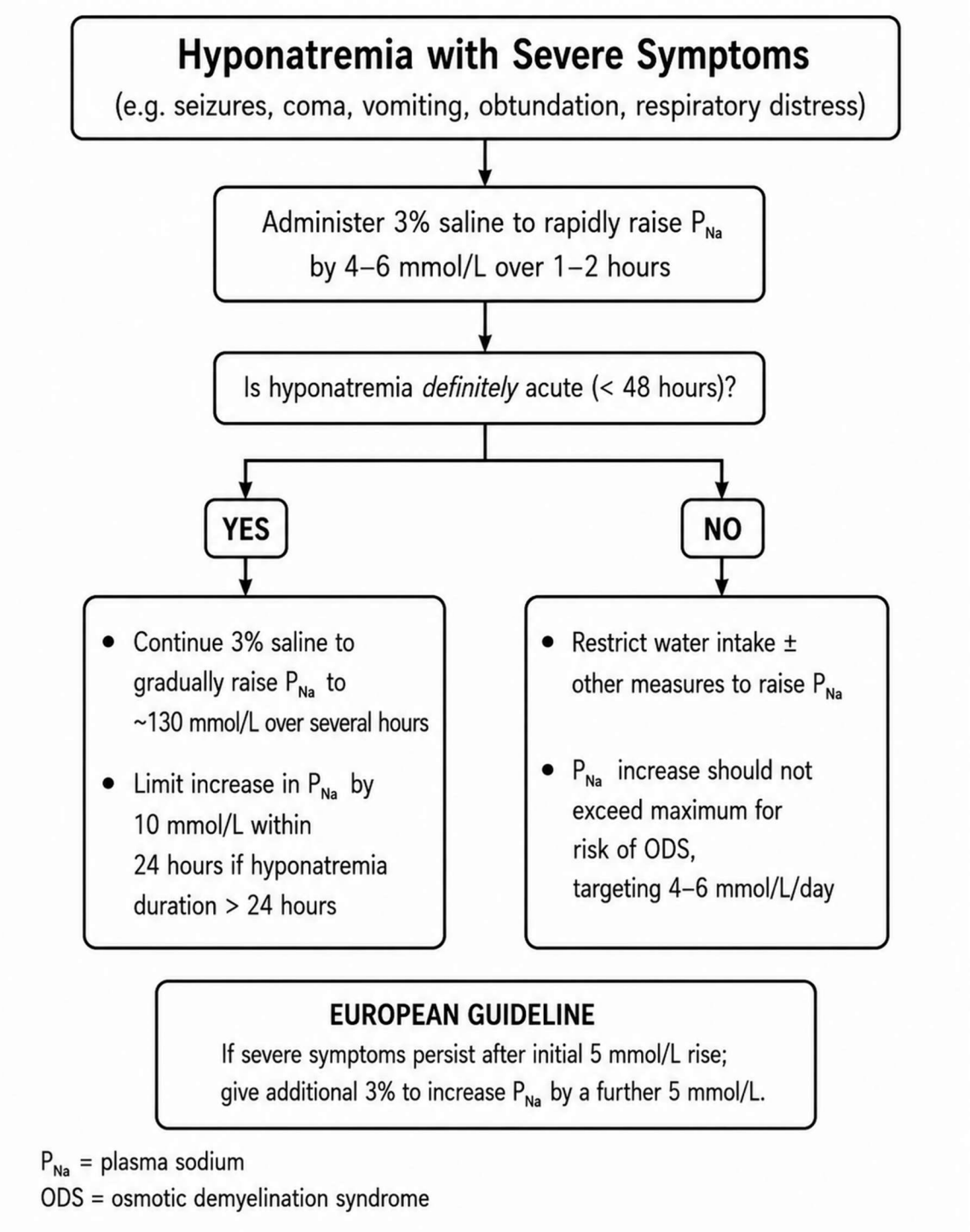
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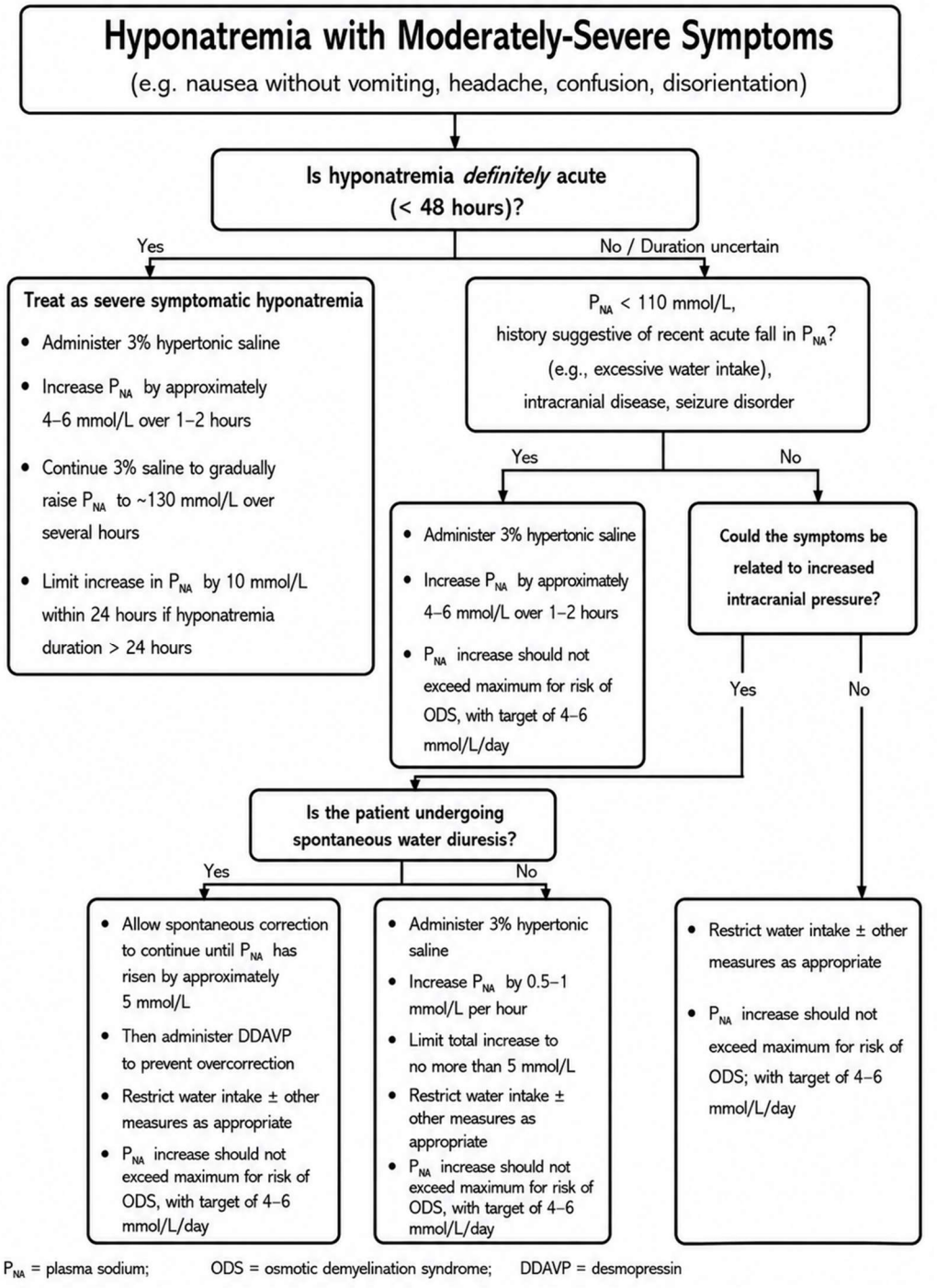


Flow Chart 1: Management of Hyponatremia with Severe Symptoms

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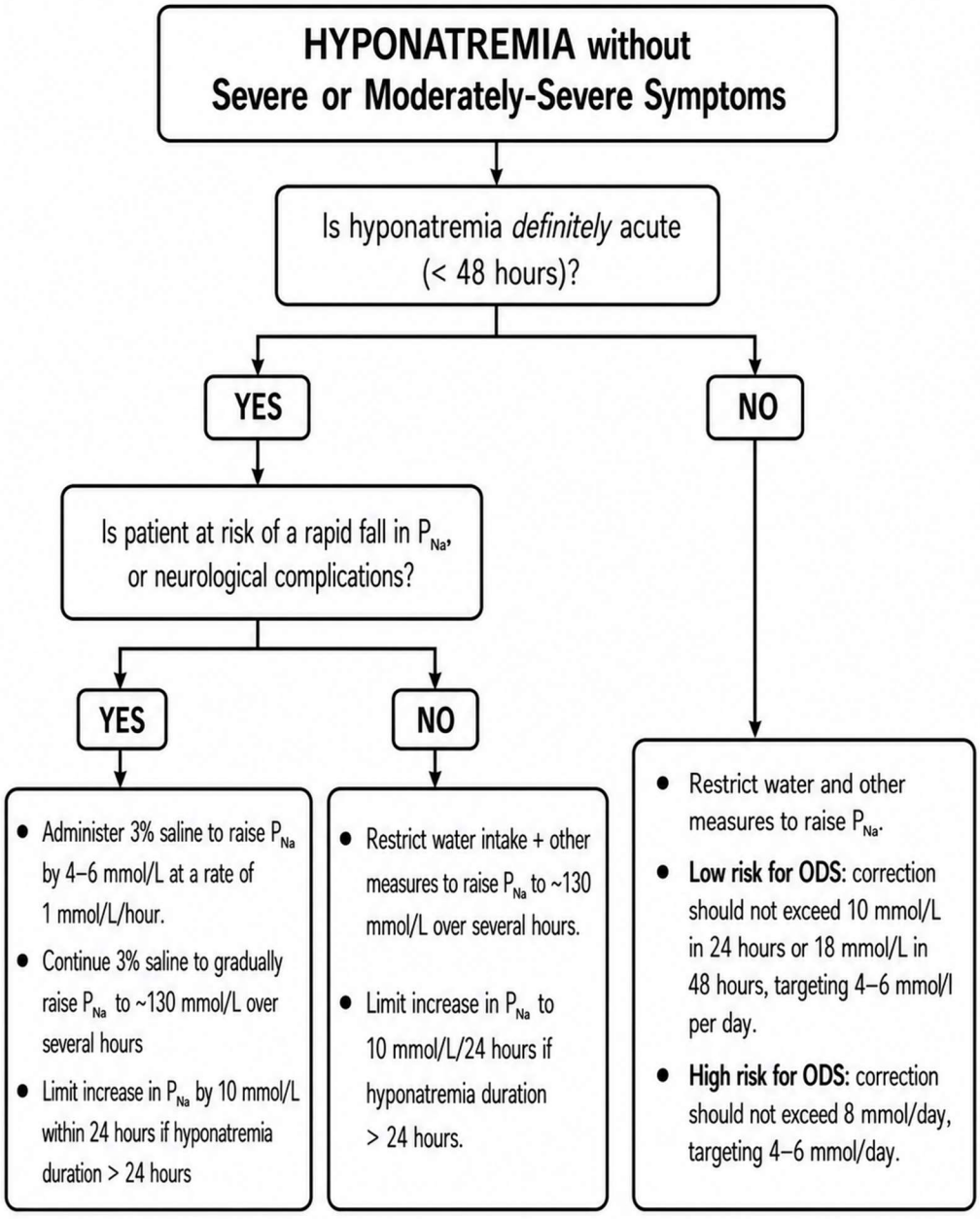


Flow Chart 2: Management of Hyponatremia with Moderately Severe Symptoms

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Flow Chart 3: Management of Hyponatremia with No Severe or Moderately Severe Symptoms

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